ENS161 (Engineering Statics) *Midterm Exam*
A.Y. 2025-2026, S1

Name ____
ID no. ____
Section ____

General instructions. Fill in the information asked above. Write legibly your solutions and answers in the answer sheets provided. Submit this questionnaire along with the answer sheet.

Part I. For fill-in-the-blank questions, each blank is worth a point. For true-or-false questions, write only *T* or *F*. For identification questions, each desired item is worth a point.

1. The ____ of a rigid body represents it as having being isolated from its surroundings, replacing physical supports with forces and couple moments.
2. *True or False.* An ideal truss does not have three-force members.
3. For 3D systems, the internal moment has two components: _____ and _____, whose axes are tangent and normal to the cross section, respectively.
4. A body is said to be ____ constrained if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.
5. *True or False.* A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate.
6. For a simply supported thirteen-member plane truss to be considered simple, it must have _____ joints.
7. *True or False.* A frictionless ball-and-socket joint resists translations and rotations.
8. A body is considered _____ constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.
9. *True or False.* A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.
10. For internal loadings of 2D systems, the ______ force and the ______ force are normal and tangent to the cross section, respectively.

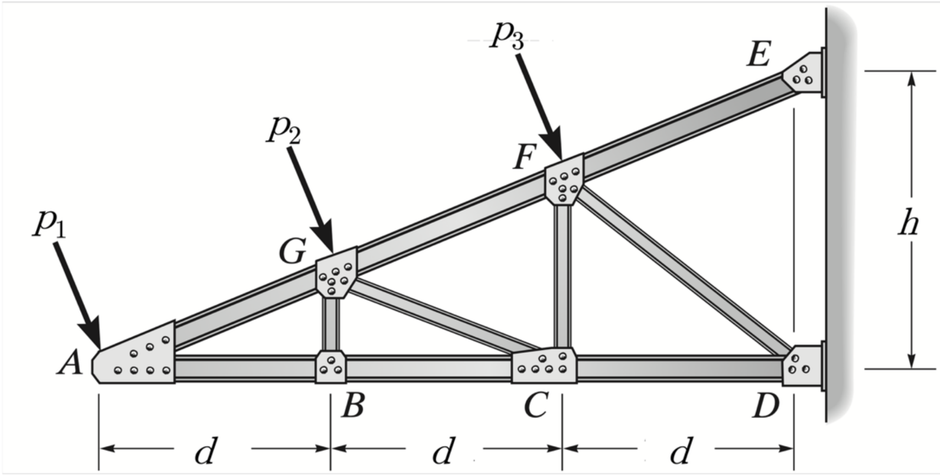
Scenario 1. A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.

11. *True or False.* This scenario is statically determinate.
12. The support reactions have magnitudes of _____ kN and _____ kN-m.
13. The maximum bending moment developed is _____ kN-m in magnitude and occurs _____ m from the free end.
14. The maximum shear force developed is _____ kN-m in magnitude and occurs _____ m from the free end.
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree _____.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree _____.

Scenario 2. A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1 m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.

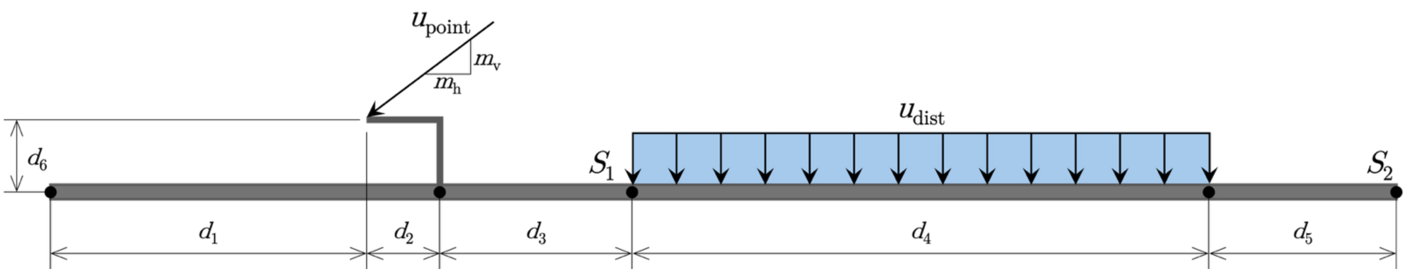
17. The roller support develops a force of _____ kN.
18. The pin support develops a force of _____ kN.
19. The bending moment developed at the right end is _____ kN-m.
20. The shear force developed just to the left of the pin is _____ kN.
21. The shear force developed just to the right of the pin is _____ kN.
22. *True or False.* There is zero shear force developed at a point 0.9 from the right end.
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree _____.
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree _____.

Part II. Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.



Problem 1. Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of p_1, p_2, and p_3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.

a. (15 points) Determine the forces developed in members *FG*, *CF*, and *CD*. Use the method of sections, making sure to indicate the section cut and show the corresponding free-body diagram.
b. (10 points) Determine the forces developed in members *DF* and *EF*. Use the method of joints, making sure to show free-body diagrams at the joints considered. *Hint:* Use the computed forces in members *FG*, *CF*, and *CD*.
c. (5 points) If the truss is to be idealized as simply supported by a pin and a roller, at which joints must the pin and the roller be placed to maintain stability of the loaded truss? Justify your answer.
d. (4 points) A contingency is considered wherein joint *A* is effectively unloaded. Identify zero-force members, if there be any, for this case.
e. (6 points) Determine the forces developed in the supports at *E* and at *D*. *Hint:* Use the computed forces in members *CD*, *DF*, and *EF*.



Problem 2. A custom beam design places supports at S_1 and at S_2, in anticipation of a concentrated load of u_{point} kips and a uniformly distributed load of intensity u_{dist} kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x, where x is the distance from the left end of the beam. Consider the case when u_{point} is 5 kips and u_{dist} is 2 kip/ft. Set the respective values of d_1, d_2, d_3, d_4, d_5, d_6 to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of m_{v} and m_{h} to 3 and 4. For static determinacy, assume a pin at S_1 and a roller at S_2.

a. (6 points) Determine the support reactions.
b. (1 point) Determine the normal force at $d_1 + d_2$ from the left end.
c. (2 points) Determine the shear force at $d_1 + d_2$ from the left end.
d. (3 points) Determine the bending moment at $d_1 + d_2$ from the left end.
e. (8 points) Determine the shear force as a function of x.
f. (8 points) Determine the bending moment as a function of x.
g. (6 points) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.
h. (6 points) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.